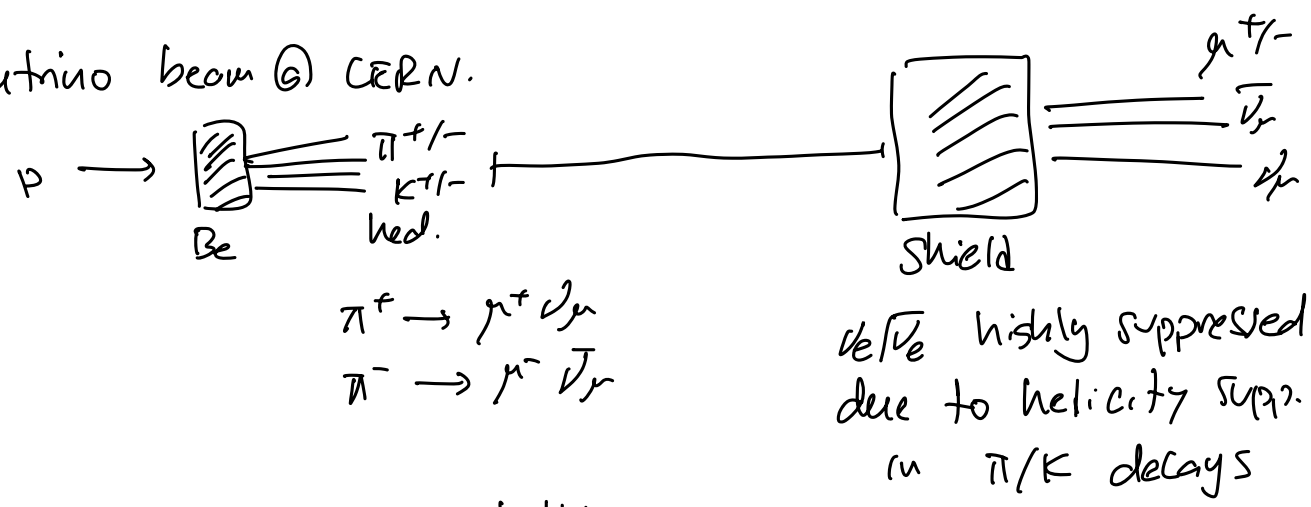


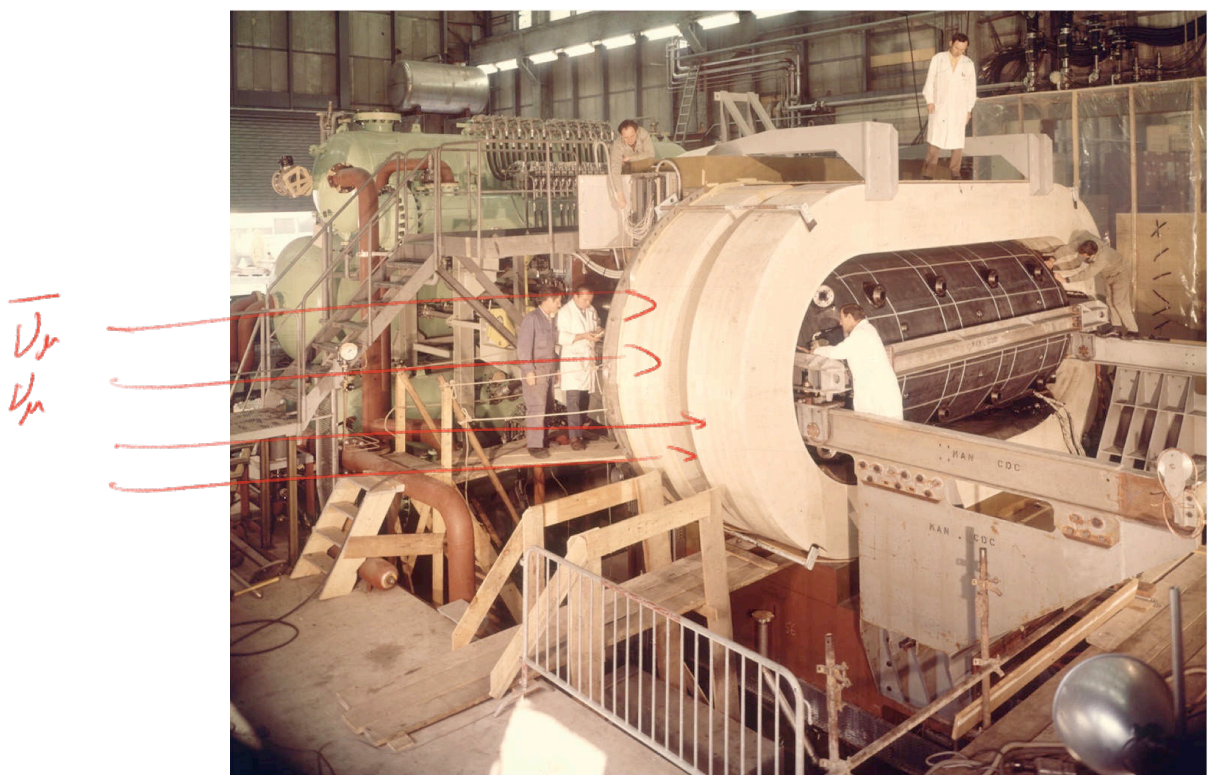
Neutral weak Current

$$\bar{\nu}_\mu / \nu_\mu + \begin{pmatrix} e \\ p \\ n \end{pmatrix} \longrightarrow \nu_\mu / \bar{\nu}_\mu + X$$

neutrino beam @ CERN.



Gargamelle detector: bubble chamber with magnetic field.



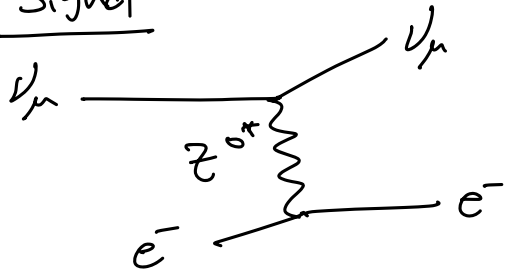
$B = 2T$ field
 A steel cylinder
 filled w/ freon
 CF_3Br (liquid)
 target (up to 100 cm)
 also used
 for bubbles



neutral current $\nu_\mu + \begin{matrix} e^- \\ p \\ n \end{matrix} \longrightarrow \nu_\mu + \begin{matrix} e^- \\ n/p \\ \text{hadrons} \end{matrix}$

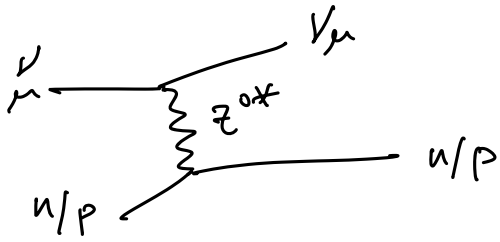
primary background process: charged weak current

Signal



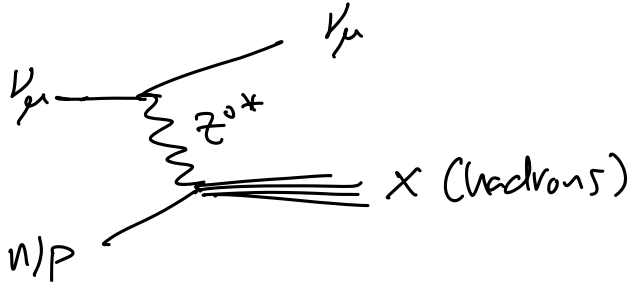
$$\nu_\mu + e^- \longrightarrow \nu_\mu + e^-$$

elastic scattering with e^-



$$\nu_\mu + n/p \longrightarrow \nu_\mu + n/p$$

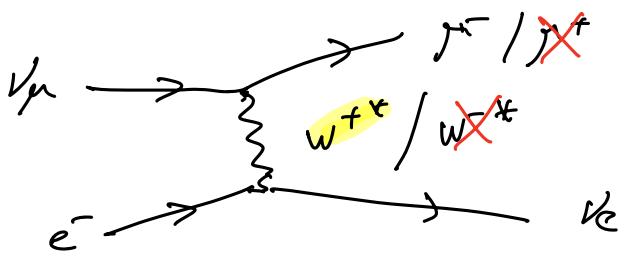
elastic scattering.



$$\nu_\mu + n/p \longrightarrow \nu_\mu + \text{hadrons}$$

at least 1 baryon.
+ mesons

Background (charge current)



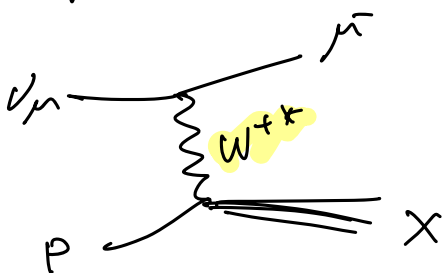
lepton # conservation

$$\nu_\mu + e^- \longrightarrow \mu^- + \nu_e$$



$$\nu_\mu + n \longrightarrow \mu^- + p/x$$

inelastic

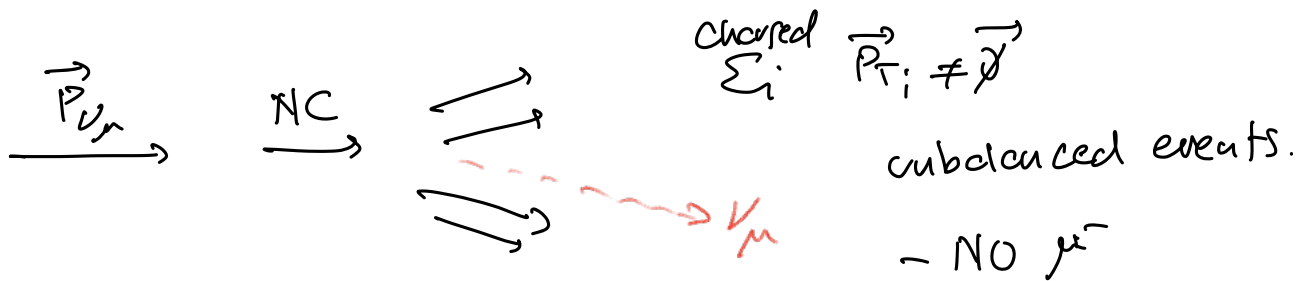
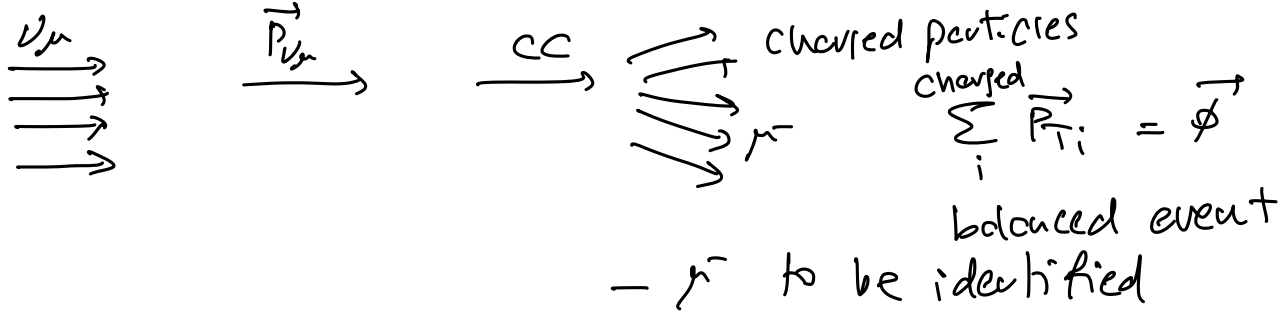


$$\nu_\mu + p \longrightarrow \mu^- + \Delta^{++} (1232)$$

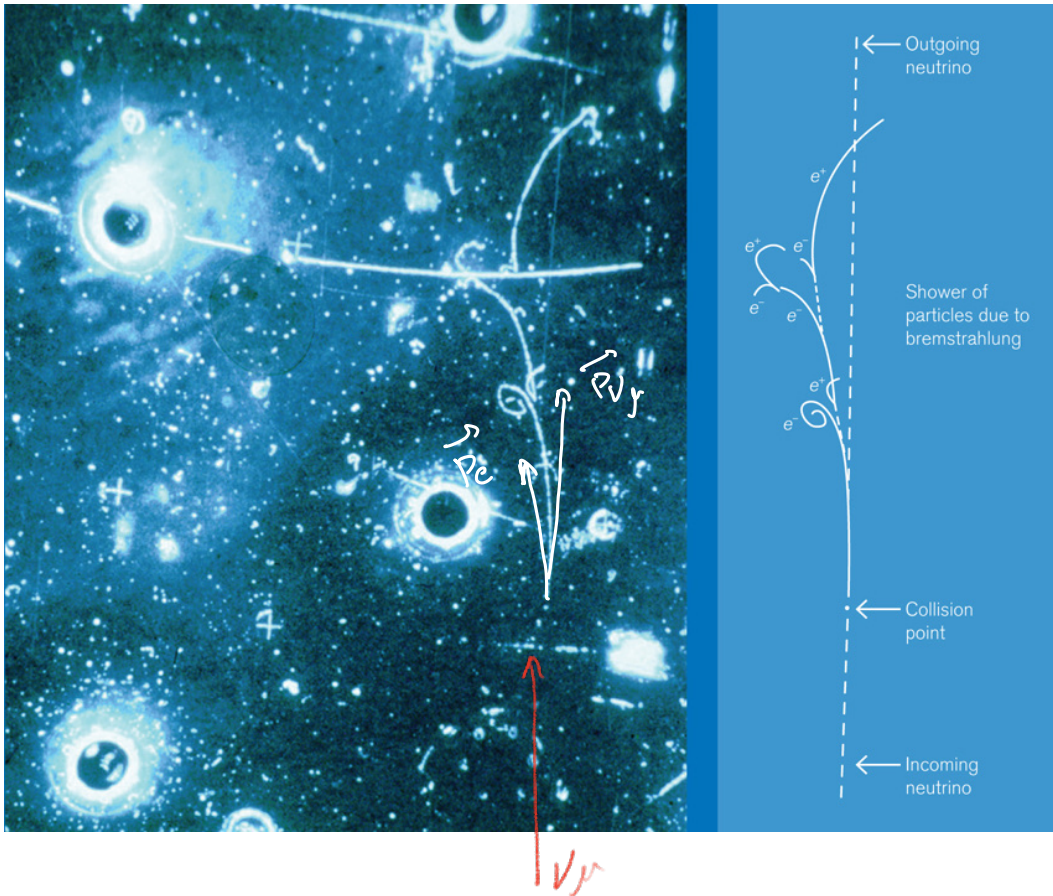
$$+ \pi^+ + \pi^+ + n$$

$$+ \text{hadrons}$$

inelastic



$\Rightarrow$ - good tracking to decide $\sum_i \vec{P}_{T,i} \neq \vec{0}$
 - μ^- identification

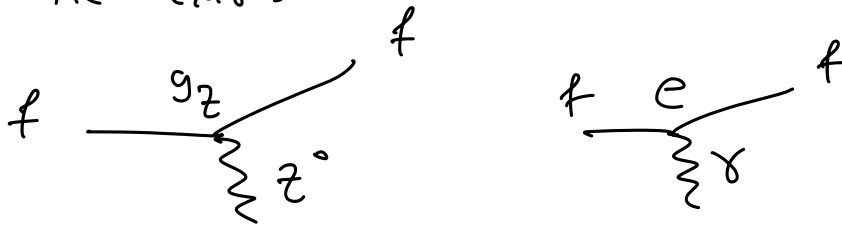


$$\frac{\#NC}{\#CC} \approx \frac{1}{3} \gg G_F^2 \Rightarrow \left. \begin{array}{l} \text{not compatible} \\ \text{with higher order} \\ \text{CC only.} \end{array} \right\}$$

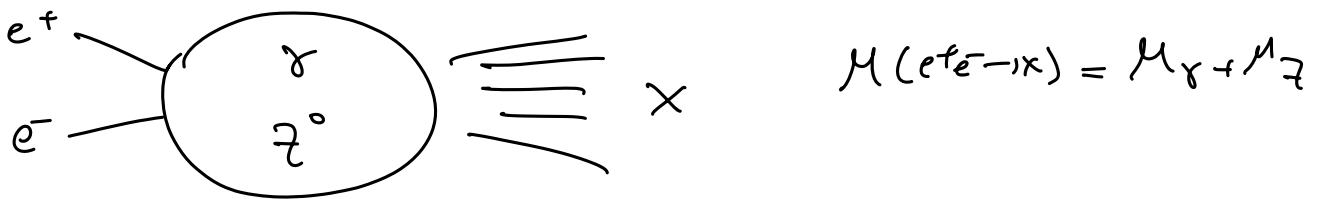
$\Rightarrow$ Indirect evidence Z^0 exists.

$\Rightarrow$ Need direct evidence for $W^\pm, Z^0$
produce them on shell.

if NC exists.



1961: Glashow propose unification of weak and EM



$$\mathcal{M}(e^+e^- \rightarrow \gamma Z) = \mathcal{M}_\gamma + \mathcal{M}_Z$$

$$\# \text{events} \propto \sigma \propto |\mathcal{M}|^2 = |\mathcal{M}_Z + \mathcal{M}_\gamma|^2$$

$$= |\mathcal{M}_Z|^2 + |\mathcal{M}_\gamma|^2 + \underbrace{\mathcal{M}_Z^\dagger \mathcal{M}_\gamma + \mathcal{M}_\gamma^\dagger \mathcal{M}_Z}_{\text{interference.}}$$

1967: Weinberg - Salam proposed unification of
weak + EM as spontaneous symm. breaking
of $SU(2)_L \times U(1)_Y$
 $\underbrace{\hspace{1cm}}$ left handed weak interact
 $\underbrace{\hspace{1cm}}$ EM

1971: 't Hooft proved that such theory is renormalizable.

$\Rightarrow$ predictions to be compared with exp. data

Glashow - Weinberg - Salam theory of EW interaction (GWS)

Left diagram: $f_i \rightarrow f_j$ via W^0 exchange. Vertex factor: $ig_W \gamma^\mu (1 - \gamma^5)$.

Right diagram: $f_i \rightarrow f_i$ via Z^0 exchange. Vertex factor: $ig_Z \gamma^\mu (C_V - C_A \gamma^5)$.

Legend for f_i : $e^\pm, \mu^\pm, \tau^\pm, \nu_i, q_i, \bar{q}_i$.

In the GWS theory C_V, C_A are exactly predicted for all fermions

f	c_V	c_A
ν_e, ν_μ, ν_τ	$\frac{1}{2}$	$\frac{1}{2}$
e^-, μ^-, τ^-	$-\frac{1}{2} + 2\sin^2\theta_W$	$-\frac{1}{2}$
u, c, t	$\frac{1}{2} - \frac{4}{3}\sin^2\theta_W$	$\frac{1}{2}$
d, s, b	$-\frac{1}{2} + \frac{2}{3}\sin^2\theta_W$	$-\frac{1}{2}$

θ_w : weak mixing angle

$$\sin^2 \hat{\theta}_W(M_Z) = 0.23122 \pm 0.00015$$

theory: $SU(2)_L \times U(1)_Y$ Gauge groups

3 bosons W_1, W_2, W_3 1 boson B

$$w_1, w_2 \rightarrow (w_1 \pm i w_2) \equiv W^\pm \text{ physical charge vector bosons (massive)}$$

$$\begin{pmatrix} z^0 \\ A \end{pmatrix} = \begin{pmatrix} \cos \theta_W & -\sin \theta_W \\ \sin \theta_W & \cos \theta_W \end{pmatrix} \begin{pmatrix} W_3 \\ B \end{pmatrix}$$

$$m_W = m_Z \cos \theta_W$$

physical
States

2°: massive

γ : massless

classical photon

A: γ field

$$g_e \equiv e$$

$$g_w = \frac{e}{\sin \theta_w}$$

$$g_z = \frac{e}{\sin \omega \cos \omega}$$

$$e = g \sin \theta_w = g' \cos \theta_w$$

δ : coupling in $SU(2)_L$
 g' : coupling in $U(1)_Y$

Exp test of theory

1) measure rates of events $\propto g_W^2, g_Z^2$

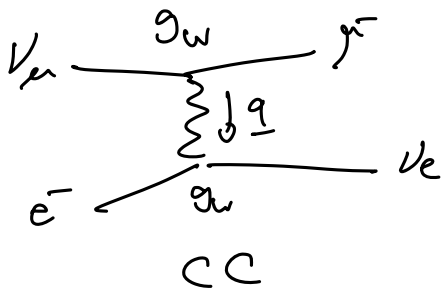
$\Rightarrow$ compute/extract C_V, C_A

$\Rightarrow$ compare to theory prediction.

2) produce W, Z on shell.

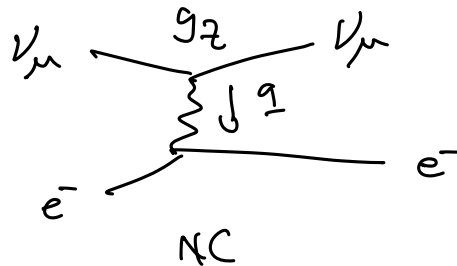
— measure their mass $\Rightarrow$ verify $m_W = m_Z \cdot \cos \theta_W$

1) measure C_V, C_A for neutral current



$$\mu_{CC} \propto g_W^2$$

$$\nu_\mu + e^- \rightarrow \mu^- + \nu_e$$



$$\mu_{NC} \propto g_Z^2$$

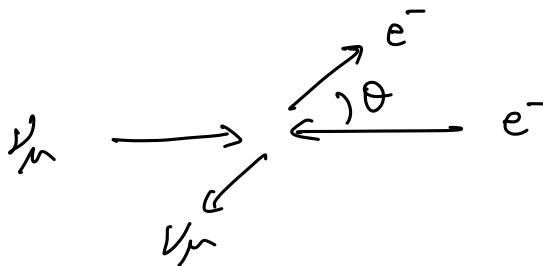
$$\nu_\mu + e^- \rightarrow \nu_\mu + e^-$$

$$\sigma_{CC}(\nu_\mu + e^- \rightarrow \mu^- + \nu_e) = \sigma(g_W) = \sigma(e, \sin \theta_W)$$

$$\sigma_{NC}(\nu_\mu + e^- \rightarrow \nu_\mu + e^-) = \sigma(g_Z) = \sigma(e, \sin \theta_W, \cos \theta_W)$$

in exp regime $q^2 \ll m_W^2, m_Z^2$

In center of mass reference



E : energy of ν, e in CMS.

Griffiths 9.6

$$\sigma_{NC} \propto \left(\frac{g_Z^2}{m_Z^2} \right)^2 E^2 (C_V^2 + C_A^2 + C_V C_A)$$

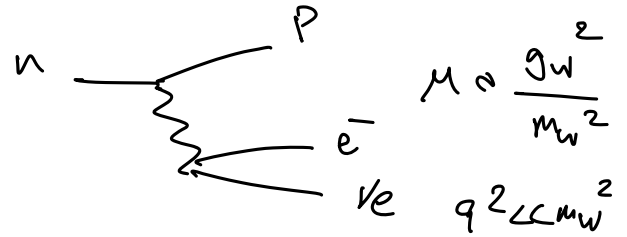
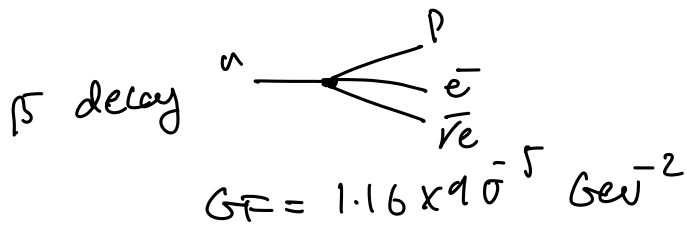
$$\frac{\sigma_{NC}}{\sigma_{CC}} = \frac{\#(\nu_\mu + e^- \rightarrow \nu_\mu + e^-)}{\#(\nu_\mu + e^- \rightarrow \mu^- + \nu_e)} = \frac{1}{4} - \sin^2 \theta_W + \frac{4}{3} \sin^4 \theta_W$$

good agreement between theory & 0.09

data & 0.11

e) production on shell of W, Z

prediction of mass from Fermi theory $\rightarrow$ GWS

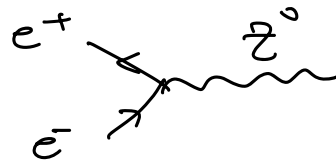
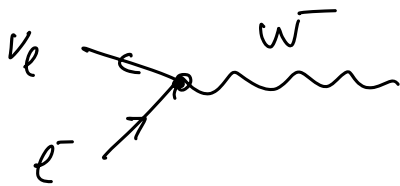


#events $\propto G_F^2$

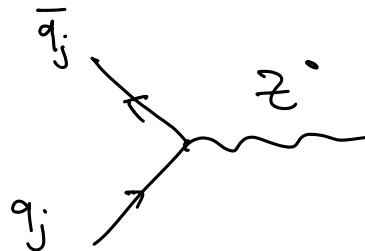
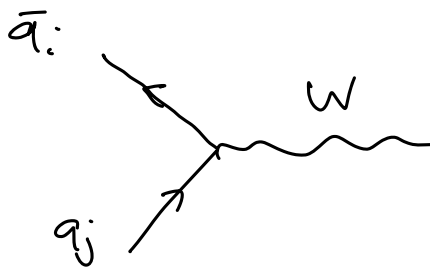
#events $\propto \left(\frac{g_W^2}{m_W^2}\right)^2$

$$G_F \approx \frac{g_W^2}{m_W^2} \Rightarrow m_W \approx 80 \text{ GeV}$$

how to produce W/Z on shell if $m \approx 80 \text{ GeV}$



not intense enough beam of $\nu_l \Rightarrow$ not feasible experimentally.



need $\bar{p}$ for $\bar{q}_j$

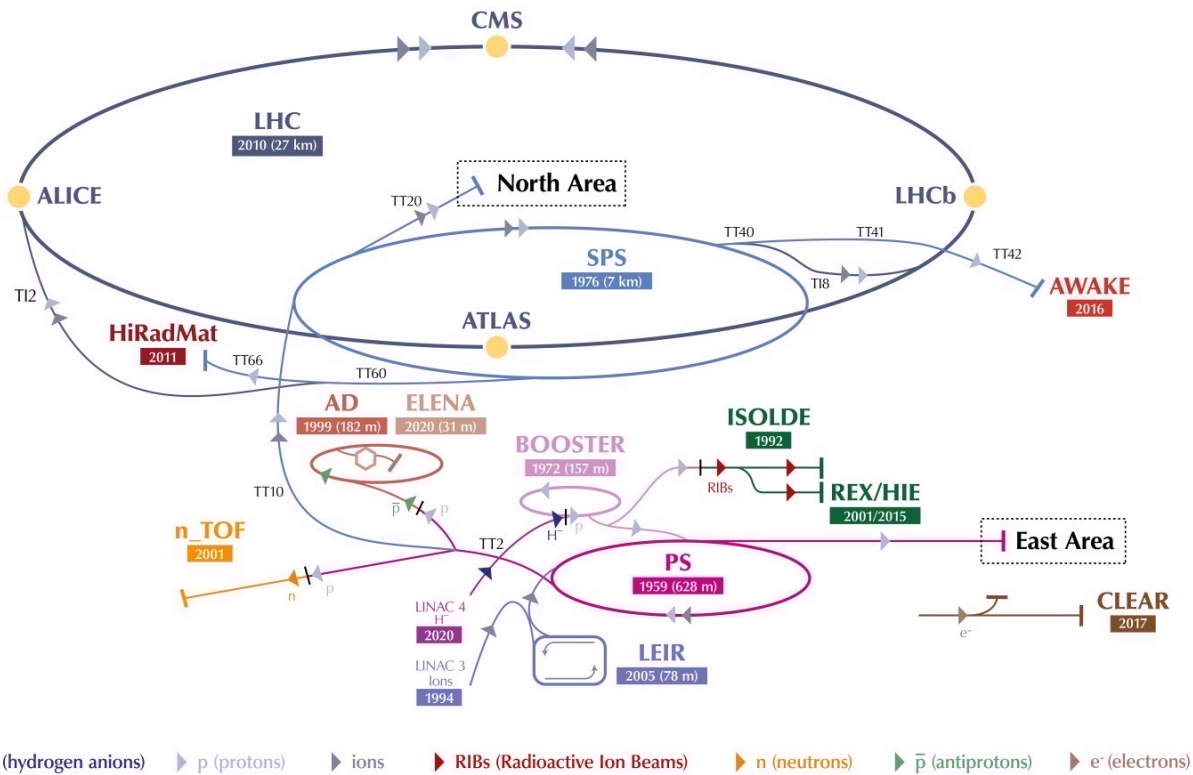
SPS @ CERN: super proton synchrotron

1976: Rubbia proposed to modify the machine and create $SP\bar{P}S$

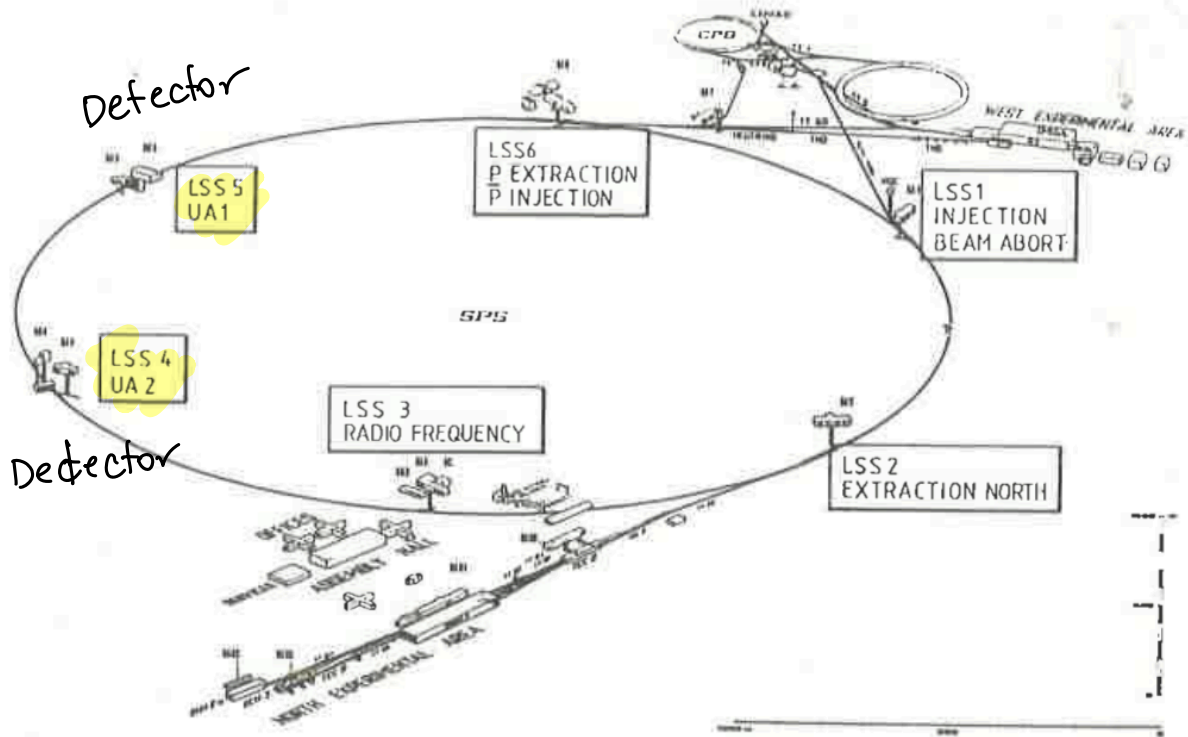
$p + \bar{p}$ collisions.

The CERN accelerator complex

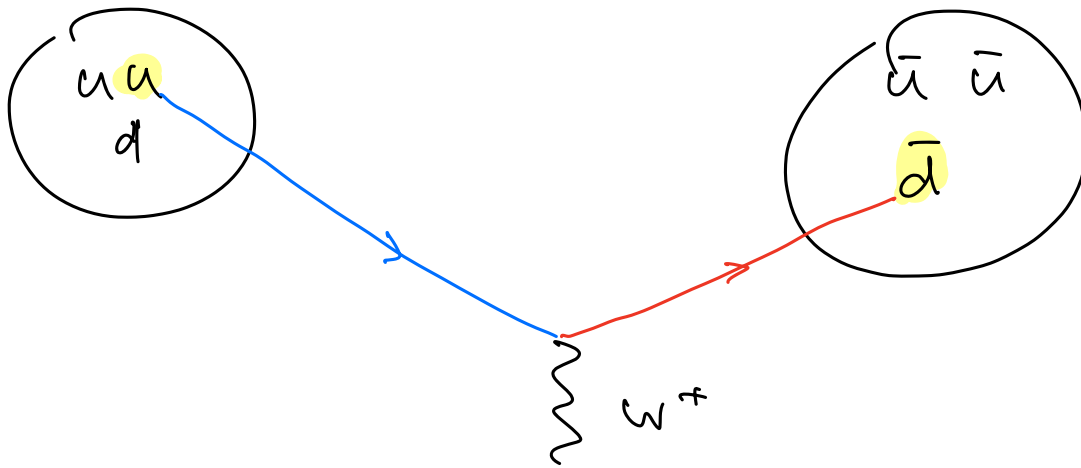
Complexe des accélérateurs du CERN



LHC - Large Hadron Collider // SPS - Super Proton Synchrotron // PS - Proton Synchrotron // AD - Antiproton Decelerator // CLEAR - CERN Linear Electron Accelerator for Research // AWAKE - Advanced WAKEfield Experiment // ISOLDE - Isotope Separator OnLine // REX/HIE - Radioactive EXperiment/High Intensity and Energy ISOLDE // LEIR - Low Energy Ion Ring // LINAC - LINEar ACcelerator // n_TOF - Neutrons Time Of Flight // HiRadMat - High-Radiation to Materials



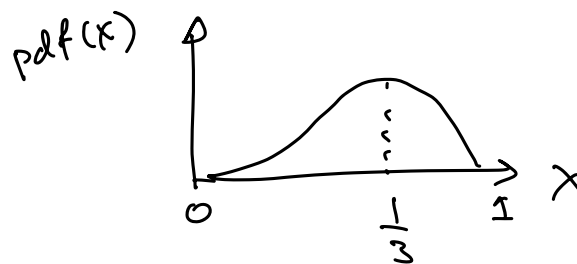
$$\underline{P}_1 = (E, 0, 0, E) \quad \xrightarrow{z} \quad \underline{P} \quad \xleftarrow{\quad} \quad \underline{P}_2 = (E, 0, 0, -E)$$



$$\underline{P}_u = x_u \underline{P}_1$$

$$\underline{P}_{\bar{d}} = x_d \underline{P}_2$$

x_u, x_d : change for every collision.



$$x \in (0, 1)$$

$$\langle x \rangle \approx \frac{1}{3}$$

$$\sqrt{s} = 2E_{\text{beam}}$$

$$S_{p\bar{p}S} \quad \sqrt{S_{p\bar{p}}} = 540 \text{ GeV}$$

$$E = 270 \text{ GeV}$$

$$\sqrt{S_{u\bar{d}}} = \sqrt{(E_u + E_{\bar{d}})^2} \approx \langle x \rangle (E + E) = \langle x \rangle \sqrt{S_{p\bar{p}}}$$

$$\Rightarrow \sqrt{S_{u\bar{d}}} \approx \frac{540}{3} \text{ GeV} \approx 180 \text{ GeV.}$$

$\Rightarrow$ probe production of W, Z for $m \approx 180 \text{ GeV}$